

DataMind AI

Machine Learning Analysis Report

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This report contains the complete analysis performed by DataMind AI.

Dataset Summary

Property	Value
Problem Type	regression
Target Column	MonthlyIncome
Recommended Model	Linear Regression
Confidence	70

Model Training Results

Rank	Model	Score
1	XGBoost	0.9131
2	Random Forest	0.8985
3	Decision Tree	0.8675
4	Linear Regression	0.8606

Best Model

Property	Value
Model	XGBoost
Score	0.9131

Cross Validation

Metric	Value
Mean Score	0.8179
Standard Deviation	0.1226

Fold	Score
Fold 1	0.7576
Fold 2	0.9508
Fold 3	0.6171
Fold 4	0.9335
Fold 5	0.8306

Hyperparameter Tuning

No hyperparameter tuning was performed.

Feature Importance

Feature	Importance
cat__ProductPitched_Basic	0.4193
cat__ProductPitched_Deluxe	0.3048
cat__ProductPitched_Standard	0.1128
cat__ProductPitched_King	0.0582
num__CustomerID	0.0198
cat__Gender_Female	0.0107
cat__MaritalStatus_Unmarried	0.0076
num__NumberOfTrips	0.0068
num__Age	0.0062
cat__ProductPitched_Super Deluxe	0.0059

Model Recommendation

Property	Value
Recommended Model	Linear Regression
Confidence	70

Recommendation Reasons

- Simple regression baseline.
- Easy to interpret coefficients.

Principal Component Analysis (PCA)

PCA analysis has not been performed.

Generated Visualizations

No visualization images were found.

Executive Summary

- Problem Type : regression
- Target Column : MonthlyIncome
- Recommended Model : Linear Regression
- Confidence : 70
- Cross Validation Mean : 0.8179
- Cross Validation Std : 0.1226

Recommendations

- Use the recommended model for deployment.
- Validate predictions using unseen data before production use.
- Consider collecting additional data if model performance is unsatisfactory.
- Regularly retrain the model as new data becomes available.
- Monitor model performance over time.

End of Report

This report was automatically generated by DataMind AI.

Generated using FastAPI, Scikit-Learn, XGBoost and ReportLab.